Delta Hedging of European Options

How rebalancing frequency, volatility misspecification and transaction costs affect hedging errors

MAC F244 · Semester I, AY 2026–27 · Project topic 12
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The seller's problem in plain English

- We sell a call: receive premium V_0 today, owe max(S_T − K, 0) at maturity
- Delta hedge: hold $\Delta = \partial C/\partial S$ shares and keep adjusting as the price moves
- Black–Scholes: continuous, costless trading with the true σ ⇒ the payoff is replicated exactly
- Reality breaks all three assumptions. We measure the leftover hedging error P&L_T



A · Discrete trading
Rebalance only N times



B · Wrong volatility
Price and hedge with σ_h ≠ σ



C · Transaction costs
Pay c per unit traded



D/E · Different assets
Volatility regimes, moneyness, maturity, real data

Model: GBM under two measures

Physical measure $\mathbb{P}$ (real world)

$$dS = \mu S\,dt + \sigma S\,dW$$

Generates the paths on which hedging P&L is realised (experiments A–D, μ = 10%)

Risk-neutral measure $\mathbb{Q}$ (pricing)

$$dS = rS\,dt + \sigma S\,dW^{\varphi}$$

Prices are $\mathbb{Q}$-expectations: $C_0 = e^{(-rT)}\,\mathbb{E}^{\varphi}[(S_T - K)^+]$. Checked: MC under $\mathbb{Q}$ gives z = −0.35; the same estimator under $\mathbb{P}$ is rejected (z = 401)

Exact simulation (no discretisation error): $S(t+\Delta t) = S(t)\cdot\exp[(m - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\cdot Z],\ Z \sim N(0,1),\ m = \mu \text{ or } r$

- BS assumptions: constant r and σ, no dividends, no costs, continuous trading, unlimited borrowing and short-selling
- Base case: $S_0 = K = 100$, T = 0.25 yr, r = 5%, σ = 20%. Money in currency units, time in years

From Itô's lemma to the hedge ratio

1 Itô $dV = (V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS})dt + \sigma S V_S\, dW$

2 Replicate $\Pi = V - V_S S$ has no dW term $\Rightarrow$ riskless $\Rightarrow d\Pi = r\Pi\, dt$

3 PDE $V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0,\ V(T,S) = (S-K)^+$ (μ cancels)

4 Solution $C = S\cdot N(d_1) - K\cdot e^{(-r\tau)}\cdot N(d_2),\ \ \Delta = N(d_1),\ \ \Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$

- $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau] / (\sigma\sqrt{\tau}),\ \ \tau = T - t$
- At maturity $\tau = 0$: $C = (S-K)^+$ and $\Delta = 1\{S > K\}$, coded explicitly; we never rebalance at T
- Near expiry at the money $\Gamma \sim 1/\sqrt{\tau}$ blows up, which is where discrete hedging hurts most

The seller's books: discrete self-financing hedge

t_0

$$B_0 = V_0 - \varphi_0 S_0 - c|\varphi_0| S_0$$

premium in, buy $\varphi_0 = \Delta_0$ shares

t_k

$$B_k = B_{k-1} e^{r\Delta t} - (\varphi_k - \varphi_{k-1}) S_k - c|\varphi_k - \varphi_{k-1}| S_k$$

interest, then rebalance

T

$$P\&L_T = B_{N-1} e^{r\Delta t} + \varphi_{N-1} S_T (1 - c) - (S_T - K)^+$$

sell shares, pay cash payoff

Conventions. $P\&L_T > 0 \Rightarrow$ seller gains (currency at T). Cost c per side on every trade incl. the initial purchase and final sale. Δ ignores c, so $P\&L(c) = P\&L(0) - c \cdot TV$ exactly: one simulation serves all cost levels. Check: an independent gains-form formula agrees to 10^{-13}.

Why discrete hedging cannot be exact

One-step error: $\varepsilon_k \approx \tfrac{1}{2}\Gamma_k S_k^2 \sigma^2 \Delta t \cdot (1 - Z_k^2), \quad \mathbb{E}[1 - Z^2] = 0, \quad \mathrm{Var}[1 - Z^2] = 2$

No bias
Each step's error has mean ≈ 0

Noise ∝ 1/√N
N independent errors, each of size ~T/N, so Var ~ 1/N

Continuous limit
$\Sigma(\Delta S/S)^2 \to \sigma^2 T$ (quadratic variation) ⇒ error → 0

- Kamal–Derman (ATM): sd(P&L) ≈ √(π/4) · vega · σ / √N → 0.439 at N = 63 (simulated 0.434)
- Wrong volatility σ_h: P&L ≈ ½∫ e^(r(T−t)) Γ_h S² (σ_h² − σ²) dt: a bias that does not shrink with N
- Leland: hedge and price at $\sigma_L^2 = \sigma^2[1 + \sqrt{(2/\pi)}\cdot k/(\sigma\sqrt{\Delta t})]$, k = 2c round trip

Code mirrors the maths, and it is checked

Equation	Python function
Exact GBM step	`gbm.simulate_gbm`
C, Δ, Γ, vega; τ = 0	`black_scholes.py`
Books B_0, B_k, P&L_T	`hedge.hedge_short_call`
Gains form (check)	`hedge.pnl_gains_form`
Wrong-σ formula	`hedge.predicted_pnl_misspecified`
Stats + Monte Carlo SEs	`stats.summarize`

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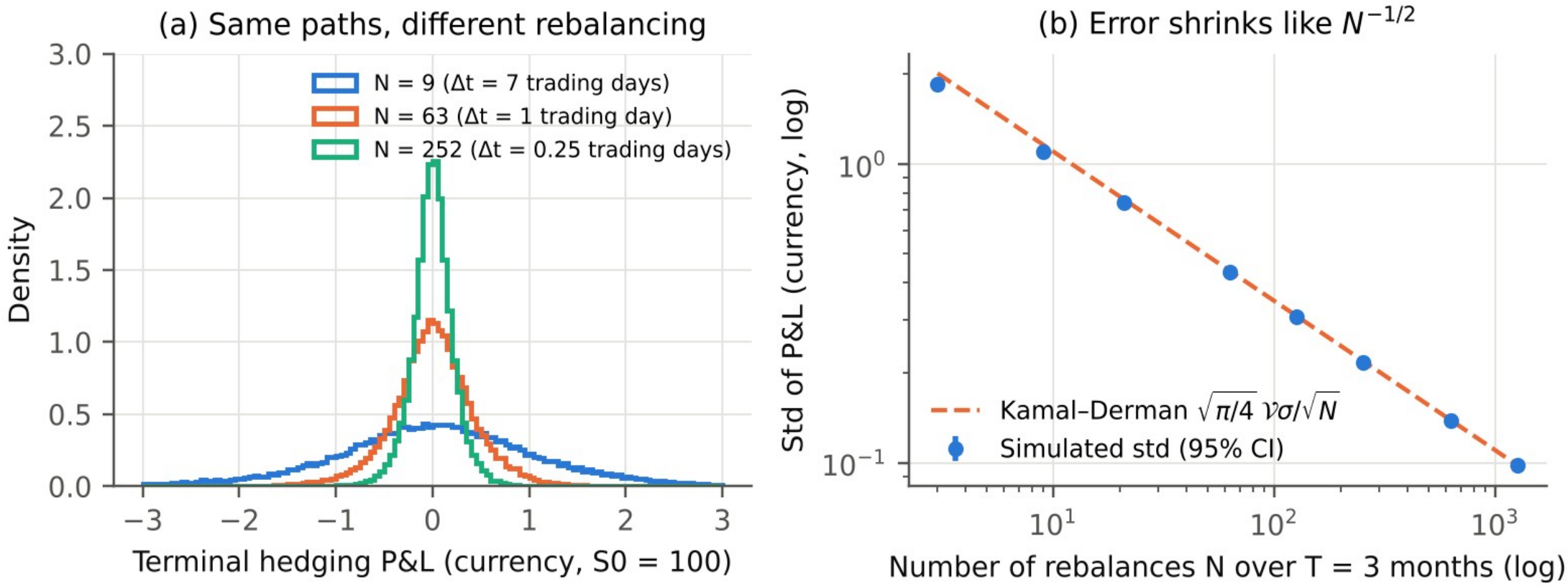
validation checks pass

- Put–call parity error 2×10^{-14}; delta vs finite difference 2×10^{-10}
- Hand ledger = simulator = gains form
- Boundaries: τ→0, deep ITM/OTM, σ→0, 0 ≤ Δ ≤ 1
- Refinement slope of error −0.49 (theory −½); two seeds agree

Common random numbers: every N uses the same 40,000 paths (fine grid, 1260 steps), so comparisons are paired. Quick mode ≈ 10 s; full mode ≈ 70 s.

A

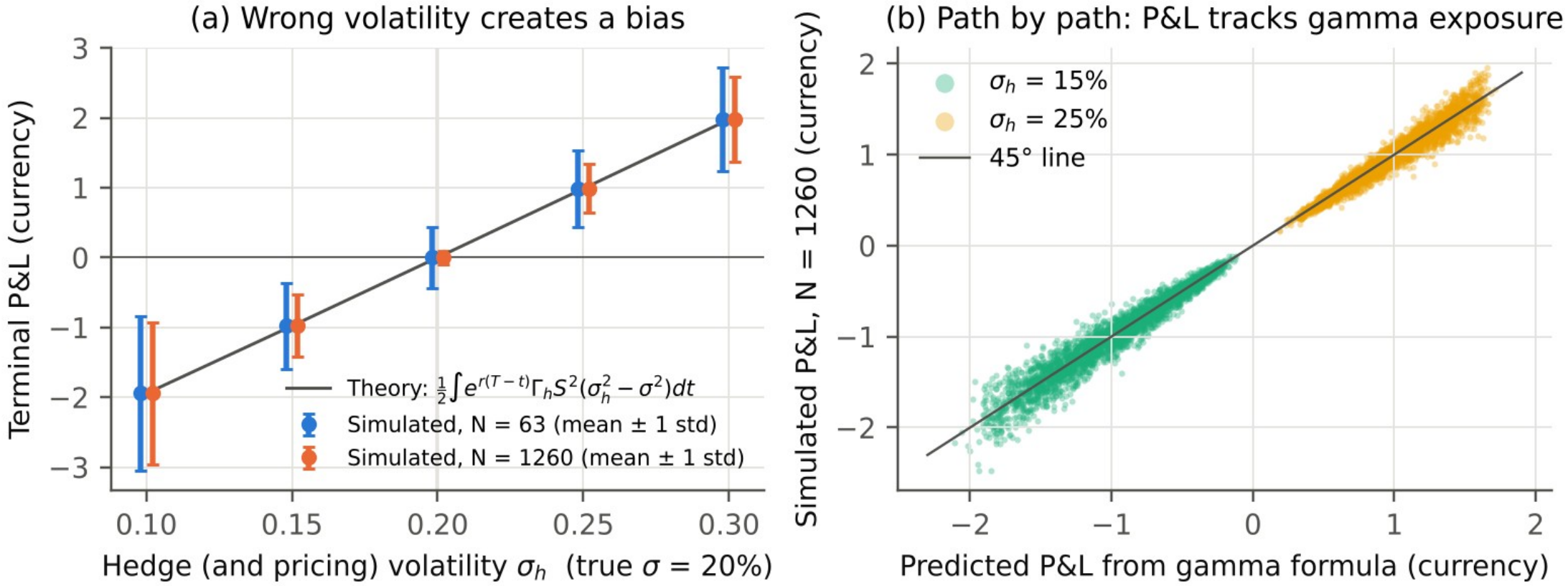
More frequent trading: error falls like 1/√N



sd 1.86 (N = 3) → 0.43 (daily) → 0.10 (N = 1260); slope −0.488 ± 0.001; mean ≈ 0. Daily error = 9% of the 4.62 premium

B

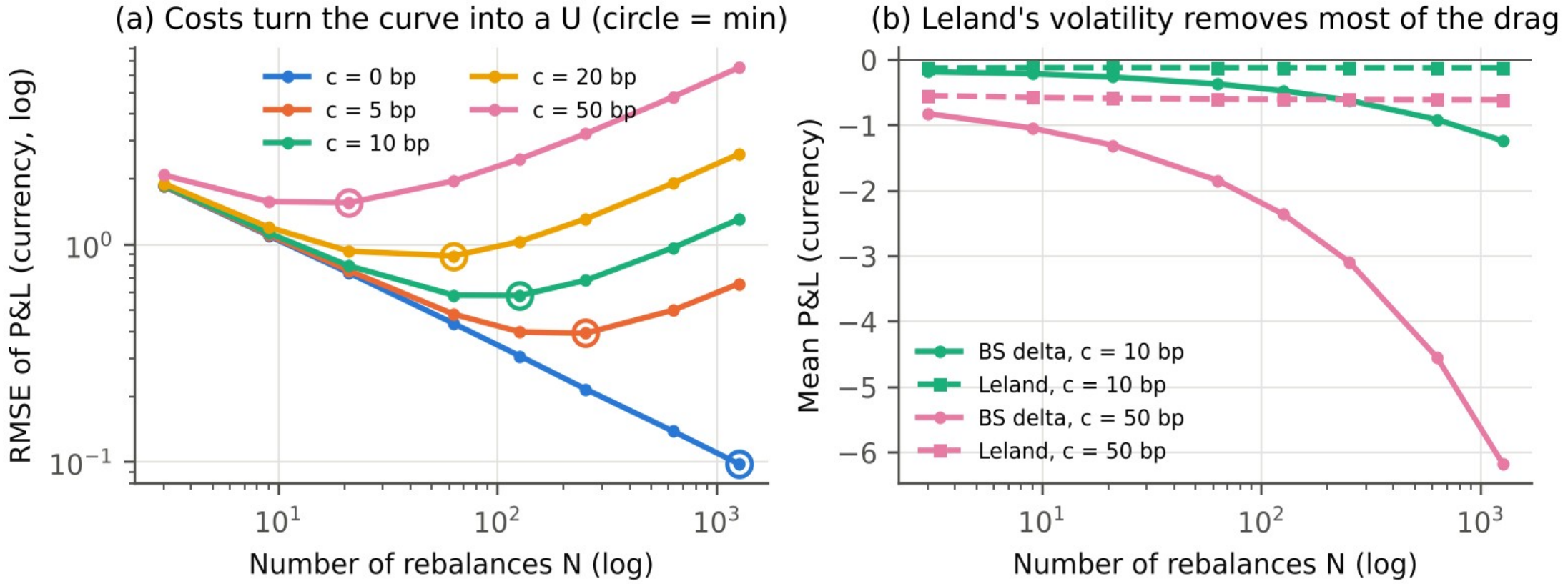
Wrong volatility: a bias that trading cannot remove



Mean P&L −1.94 / −0.98 / 0 / +0.99 / +1.98 for σ_h = 10–30%, matching the gamma formula, at N = 63 and at N = 1260

C

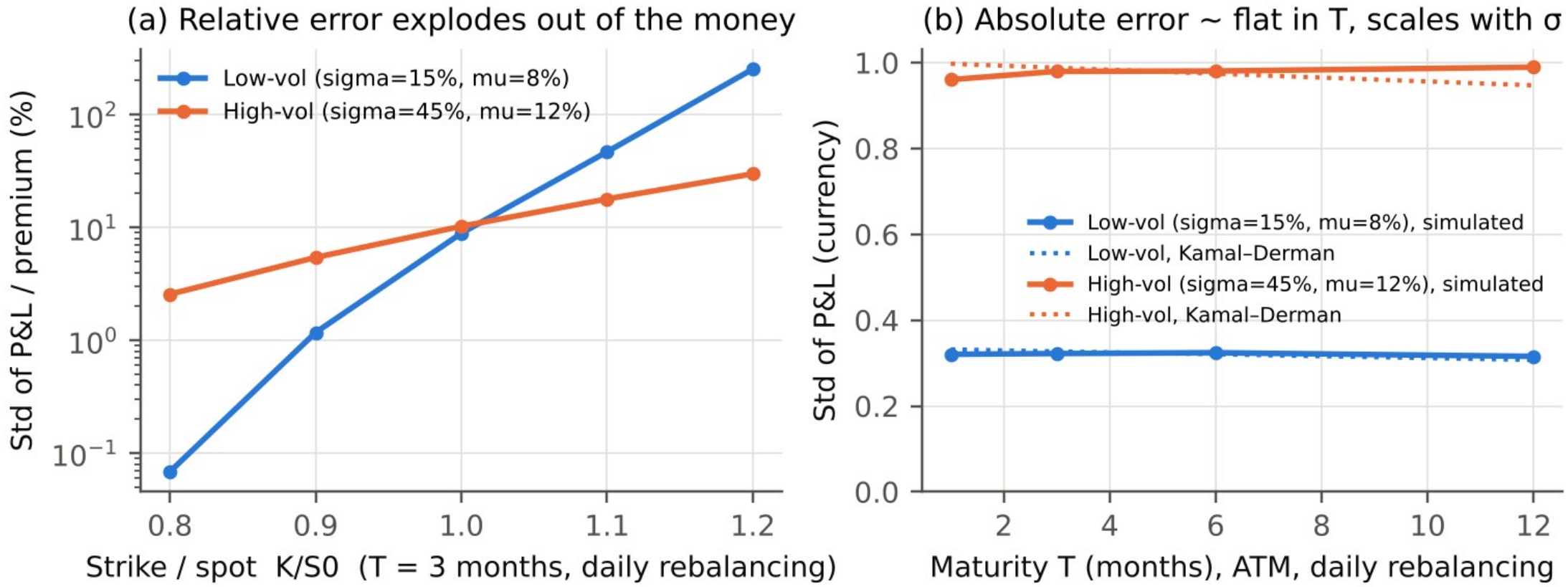
Costs: there is a best rebalancing frequency



RMSE-optimal N: 1260 / 252 / 126 / 63 / 21 for 0 / 5 / 10 / 20 / 50 bp. Leland's σ_L leaves only the 0.12 setup + unwind cost (10 bp)

D

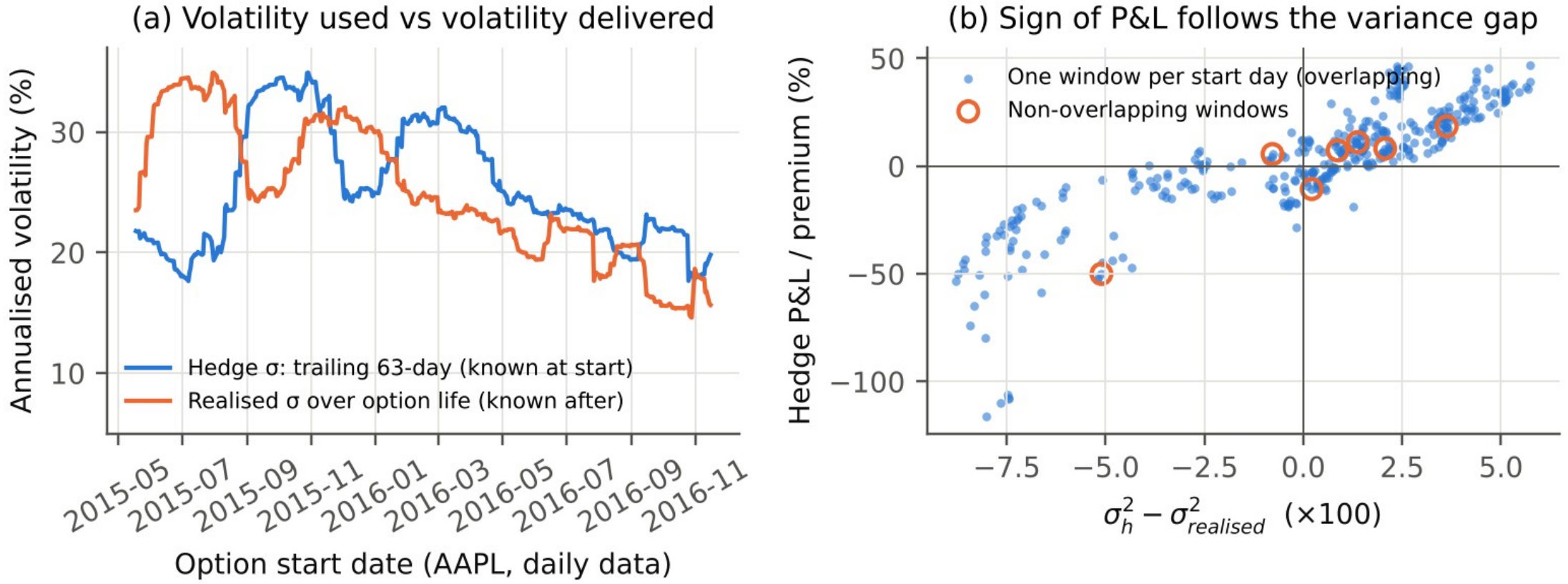
Asset regimes, moneyness and maturity



ATM daily-hedge sd 0.32 (σ = 15%) vs 0.98 (σ = 45%): ∝ σ. OTM $K/S_0 = 1.2$: sd = 254% of a 0.03 premium (low vol). Absolute error flat in T

E

Real prices (secondary source, illustration only)



σ̂: IBM 29%, MSFT 34%, GOOG 39%, AAPL 55%, AMZN 59% (2000–10, monthly); AAPL 24% (2015–17, daily). Backtest corr(P&L, σ_h² − σ²_real) = 0.87

What we learned

Frequency	Unbiased noise; sd ≈ $\sqrt{(\pi/4)}\cdot\text{vega}\cdot\sigma/\sqrt{N}$. 4× trading halves the error
Volatility	Bias = gamma-weighted $(\sigma_h^2 - \sigma^2)$. More trading cannot remove it
Costs	U-shaped RMSE ⇒ an optimal frequency; Leland covers rebalancing costs only
Assets	Relative error largest for OTM, short-dated, low-vol options; absolute $\propto \sigma$

Limitations and next steps

Limitations

- Constant σ and r; no jumps, dividends, bid–ask dynamics or market impact
- RMSE is one criterion among many
- Monte Carlo SEs measure sampling error, not model error
- Real data: secondary source, model premiums, short overlapping sample

Next steps

- Stochastic volatility (Heston) and jumps
- Band (move-based) rebalancing instead of fixed times
- Real option quotes and implied volatility
- Gamma hedging with a second option

References, data and AI-assistance disclosure

- Black & Scholes (1973) JPE 81(3); Merton (1973) Bell J. Econ. 4(1)
- Boyle & Emanuel (1980) JFE 8(3); Leland (1985) J. Finance 40(5); Kabanov & Safarian (1997) Fin. Stoch. 1
- El Karoui, Jeanblanc-Picqué & Shreve (1998) Math. Finance 8(2); Kamal & Derman (1999) Risk
- Shreve (2004) Stochastic Calculus for Finance II; Glasserman (2004) Monte Carlo Methods in Financial Engineering

- Data: vega-datasets stocks.csv; plotly/datasets finance-charts-apple.csv (provenance and hashes in data/SOURCES.md)
- AI assistance: code, text and figures were drafted with an AI assistant (Claude) and verified by running the code; see docs/ai_disclosure.md

Thank you. Questions?